

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- No recommended Cummins® service tools required

Additional Service Items

- Feeler gauge
- Copper or aluminum vise protector
- Vise
- Universal piston ring expander.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

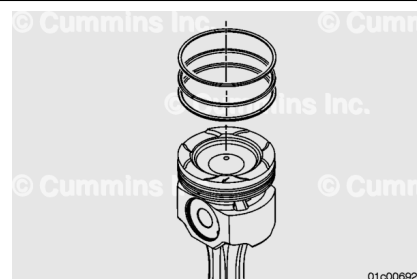
- Disconnect the batteries. See equipment manufacturer service information.
- Drain the lubricating oil system. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/1016/1016-007-037.html)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2.
(/qs3/pubsys2/xml/en/procedures/1016/1016-002-004.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-089.html)
- Remove the piston cooling nozzle. Refer to Procedure 001-046 in Section 1.
(/qs3/pubsys2/xml/en/procedures/1016/1016-001-046.html)
- Remove and disassemble the piston and connecting rod assembly. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/1016/1016-001-054.html)

Remove

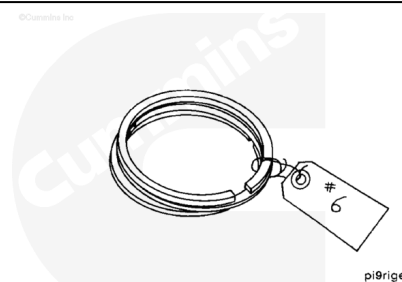
⚠ CAUTION ⚠

Do not over-expand the second piston ring.

Remove the piston rings.



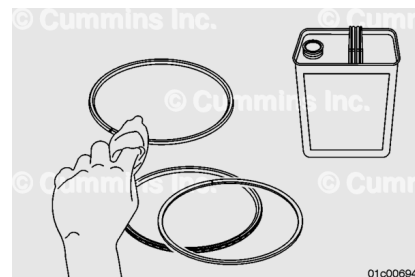
Place a tag on the piston rings. Record the cylinder number of the piston on the tag for future use, if required.



Clean and Inspect for Reuse

⚠ WARNING ⚠

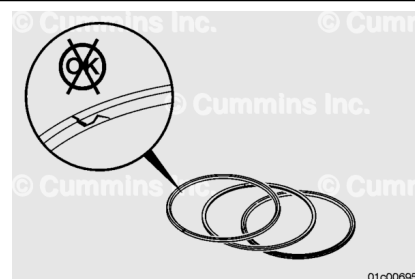
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Clean the piston rings. Use mineral spirits.

Inspect all piston rings for any cracks or chips on the ring face that mates with the liner.

Discard and replace any piston rings that are damaged.



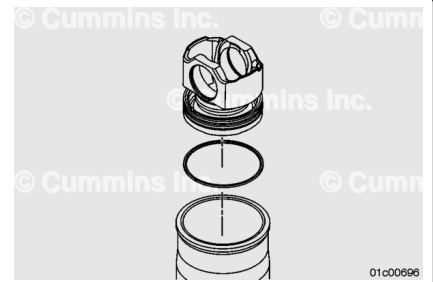
Measure

Before completing this inspection, verify the cylinder liner bore is within specification and record the cylinder liner inside diameter. Refer to Procedure 001-028 in Section 1.

(/qs3/pubsys2/xml/en/procedures/1016/1016-001-028.html)

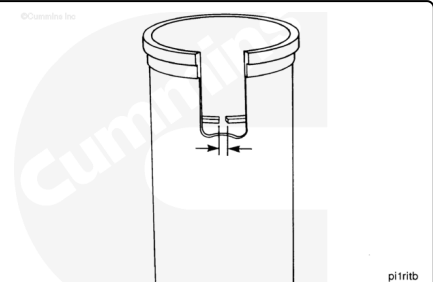
To check the piston ring gap, install the piston rings into the cylinder liner in which they are used, below the ring reversal area.

Use the top part of the piston to correctly position the piston ring in the cylinder liner.



Use a feeler gauge to inspect the ring end gap.

Piston Ring Gap			
	mm		in
Top	0.30	MIN	0.012
	0.45	MAX	0.018
Second	0.40	MIN	0.016
	0.60	MAX	0.024
Oil	0.2	MIN	0.008
	1.1	MAX	0.043



If the piston ring gaps do **not** meet specifications, the three piston rings **must** be replaced. Cummins Inc. recommends inspecting the piston and cylinder liner and to consider replacement if the piston and cylinder liner inspection does **not** meet specifications.

Install

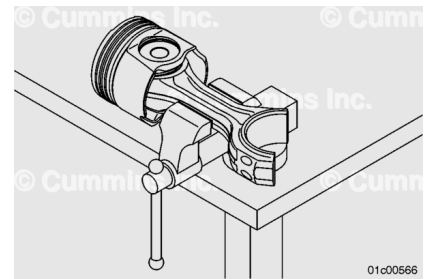
⚠ CAUTION ⚠

Use a copper or aluminum vise protector when installing the connecting rod in a vise to protect the connecting rod from damage.

When reusing piston rings, be sure the piston rings are installed on the same piston from which they are removed.

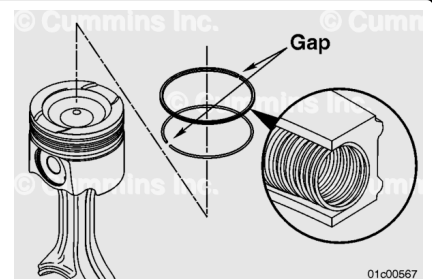
The piston ring gap of each piston ring **must not** be aligned with the piston pin or with any other ring. If the piston ring gaps are **not** positioned correctly, the piston rings will **not** seal properly.

Install the connecting rod in a vise to hold the assembly in a horizontal position while installing the piston rings.



A cross-sectioned view of an oil control ring is shown.

The two-piece oil control ring **must** be installed with the expander ring gap 180 degrees from the gap of the oil ring. Do **not** overlap the ends of the expander ring.

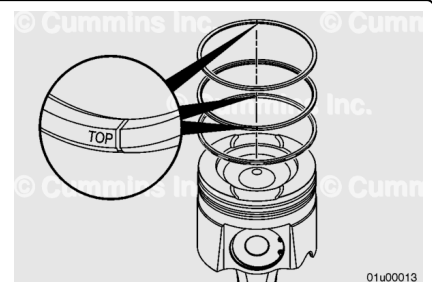


Each piston ring has the top marked for correct installation.

Install the piston rings with TOP toward the top of the piston.

Note : The oil control ring is symmetrical for this engine.

Use a universal piston ring expander to install the second ring into the piston.

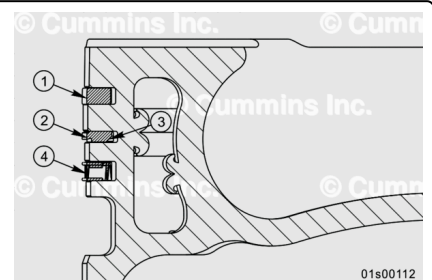


The piston **must** be held in a horizontal position to install the piston rings in order to keep the piston rings in the groove.

The top piston ring (1) is a keystone ring with no twist or cutback notch.

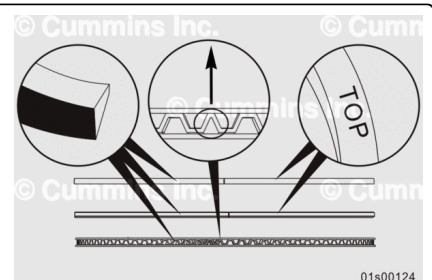
The second piston ring has a rectangular cross-section (2) with an angled cut (3) on the lower side of the outer circumference.

The oil control piston ring (4) is the bottom piston ring.



When installing new piston rings:

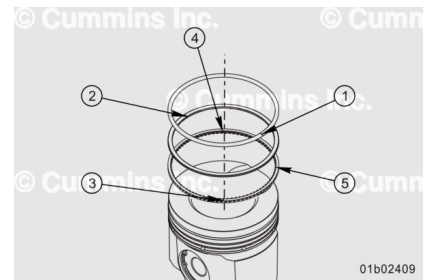
- The top piston ring will show a yellow paint stripe to the left of the gap and a "TOP 1" mark at the right of the gap.
- The second piston ring will show a green paint stripe to the left of the gap and a "TOP 2" mark at the right of the gap.



- The oil control ring consists of two pieces of oil piston rings and an expander ring. The expander ring **must** be installed with the expander joint upward. The expander ring will show a pink or blue paint stripe to the left of the expander joint.

When reusing piston rings, verify the piston rings are installed on the same piston from which the piston rings are removed.

Do **not** align the piston ring gap of each piston ring with the piston pin or with any other ring gap. The piston ring gaps **must** be at least 30 degrees apart as the picture shown. If the piston ring gaps are **not** positioned correctly, the piston rings will **not** seal properly.



1. Top piston ring gap
2. Second piston ring gap
3. Top oil ring gap
4. Bottom oil ring gap
5. Oil ring expander gap.

If a malfunction results in coolant, oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected.

Note : When installing new piston rings with used cylinder liners, higher than average blowby can be experienced for the first several hours of operation. If excessive blowby is still seen after several hours of engine operation, consult the Crankcase Gases (Blowby) Excessive troubleshooting symptom tree in Section TS in 15N CM2380 M104B Fault Code Troubleshooting Manual, Bulletin 5659765.

Note : All engines **must** be run-in after a rebuild or a repair involving the replacement of one or more piston ring sets, cylinder liners, or pistons. See the appropriate procedure in Section 14 for the correct run-in procedure.

Use the following procedure for an overview of a general run-in test. Refer to Procedure 014-999 in Section F. (</qs3/pubsys2/xml/en/procedures/1016/1016-014-999.html>)

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install and assemble the piston and connecting rod assembly. Refer to Procedure 001-054 in Section 1. (</qs3/pubsys2/xml/en/procedures/1016/1016-001-054.html>)
- Install the piston cooling nozzle. Refer to Procedure 001-046 in Section 1. (</qs3/pubsys2/xml/en/procedures/1016/1016-001-046.html>)
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/1016/1016-001-089.html>)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html>)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/1016/1016-002-004.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html>)
- Fill the lubricating oil system. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/1016/1016-007-037.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.